

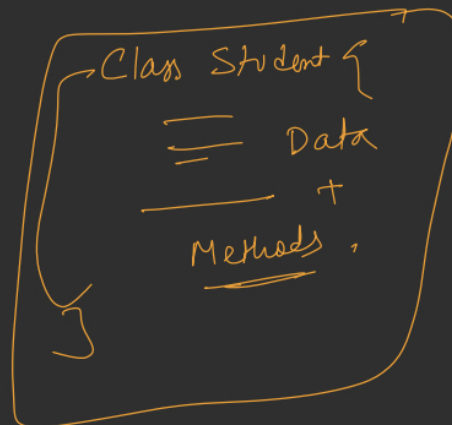
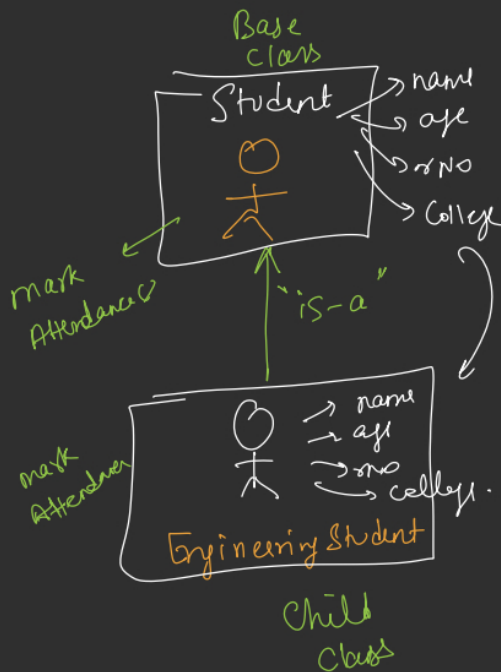
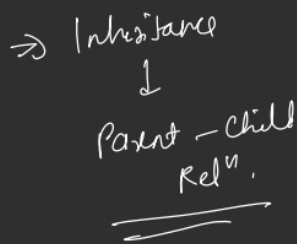
Encapsulation

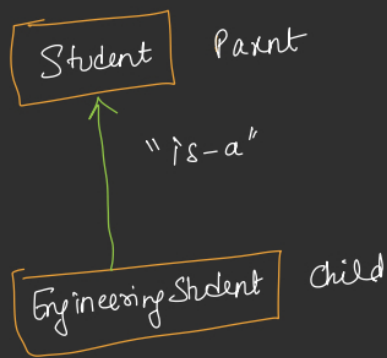


OOPS



Real
world





```

class Student {
  private:
    p1;
    p2;
  public:
};

class ES : Student {
};
  
```

Public, Private, Protected

```

class Student {
  private:
  protected:
  public:
};
  
```

Public
↳ Inheriting Student publicly;

Private
↳ Inheriting Student Privately;

```

class EngineeringStudent : Student {
};
  
```

Protected
↳ Inheriting Student Protectedly

①

Access Modifier

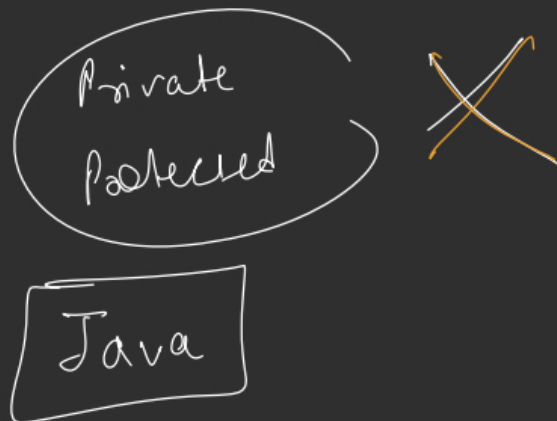
public

private

Protected

②

Publically Inherit



[C++]

Private
Protected

Library
Code

Class A {
}
}

Composition ✓

'has-a'

Class A {
 B *b;
};

Class B {
 void func();
};

A * a = new AC();
a → b → func(); (X)

Inheritance



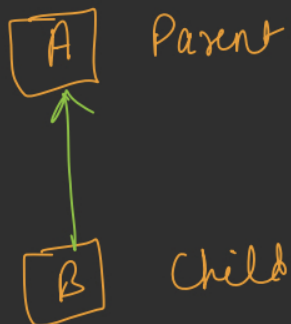
Single inheritance

Multilevel inheritance

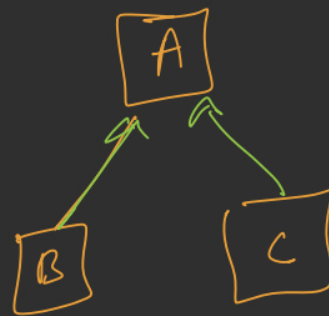
Hierarchical

Multiple X X

Single



Hierarchical



Multi-level



Multiple !

